# TRETINOIN

Teratogenic

Apply at night; photosensitivity

**CHEMICAL NAME** (2E,4E,6E,8E)-3,7-dimethyl-9-(2,6,6-trimethylcyclohex-1-en-1-yl)nona-2,4,6,8-tetraenoic acid (all-trans-retinoic acid)

**CLASS** Retinoid (topical for acne; oral for acute promyelocytic leukaemia)

**TRADE NAMES** Retin-A · Airol · Vesanoid (oral, APL) · Retacnyl

## THERAPEUTIC USES / INDICATIONS

- Acne vulgaris (comedonal and inflammatory) — topical.
- Photoageing (fine wrinkles, hyperpigmentation); topical.
- Acute promyelocytic leukaemia (APL) — oral induction (specialist).

## ADMINISTRATION

- **Topical** pea-sized amount to whole face at night on clean, dry skin (wait 20–30 min after washing).
- Avoid eyes, mouth, nostrils; use sunscreen daily.
- **Oral (APL):** capsules with food in two divided doses.

## DOSAGES

- **Acne:** 0.025–0.1% cream/gel once nightly (start low, alternate nights if irritated).
- **APL:** 45 mg/m²/day orally in 2 divided doses until complete remission (with arsenic trioxide/chemotherapy).

## PHARMACOKINETICS

- Topical: minimal absorption (<2%).
- Oral: metabolised by CYP; induces own metabolism. **t½:** ~0.7 h.

## PHARMACODYNAMICS

- Binds **retinoic acid receptors (RAR)** → normalises follicular keratinisation, comedolytic, ↑ cell turnover.
- In APL: induces differentiation of promyelocytes (PML-RARα).

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Topical: redness, peeling, dryness, burning; initial acne flare (weeks 2–4).
- Photosensitivity.
- Oral: headache, dry skin, raised lipids.

### ADVERSE EFFECTS (SERIOUS)

- Teratogenicity (oral; absolute; topical — avoid in pregnancy).
- Differentiation syndrome (APL): fever, dyspnoea, pulmonary infiltrates, weight gain; treat with dexamethasone.
- Intracranial hypertension (oral, especially with tetracyclines).

## CONTRAINDICATIONS & CAUTIONS

- Pregnancy (oral contraindicated; topical avoid) and planning pregnancy.
- Eczema, sunburn, broken skin (topical).
- Oral: with tetracyclines; hypervitaminosis A.

### KEY INTERACTIONS

- **Benzoyl peroxide (same time):** may degrade tretinoin (use separately).
- **Tetracyclines (oral tretinoin):** intracranial hypertension.
- **CYP3A4 inhibitors/inducers:** oral levels.

## NURSING MANAGEMENT

- Confirm contraception/pregnancy status.
- Topical: teach application and sun protection.
- APL: monitor for differentiation syndrome (weight, breathing, temperature), coagulation (DIC).

## PATIENT EDUCATION

- Apply a pea-sized amount at night to the whole face; acne may worsen briefly before improving.
- Use sunscreen and avoid sunbeds.
- Do not use if pregnant or trying to conceive.
- Avoid waxing treated areas.

**KEY POINTS**

- Retinoid; normalises keratinisation.
- Oral for APL — differentiation syndrome.
- Topical acne at night; photosensitivity.
- Teratogenic.

**SOURCES CHECKED** **[976]** Tretinoin 10mg soft capsules (UK SmPC)

*Full references p. 432–457*